

Writing technical papers

Manel Rodríguez Soto

July 2024

What are these notes about?

No one knows how to properly write papers at the beginning. The beginning is hard and difficult because your supervisors are constantly telling you that every single sentence you write is wrong. And then you get sad and depressed...

But, wait a minute. There's nothing wrong with that! No one actually told you how to write papers, so then it is perfectly normal that you don't write good papers. You just need to learn.

Imagine someone who has never played basketball. It's his first day doing a match, and he hasn't even trained a single exercise. What do you expect? Most likely, he will do everything wrong from beginning to end. Or not totally wrong, but with room to improve. He will need to learn the basics, train, and then come back. With writing papers it's the same thing. The key to writing good papers is simply to learn to do papers.

These notes will teach you the basics on how to structure each section of your paper, and several tips and tricks to ensure to improve your writing overall. The notes also provide some exercises to make you evaluate your own writing, but at the end of the day you need to remember: the only way to improve your writing is by:

1. Read papers and analyze how they are written.
2. Try to write your own papers and then evaluate them to assess what needs to be improved and how.
3. Repeat.

These notes are structured as follows.

- First, I will provide a brief section with very general tips that will serve for any section of the paper. These will introduce you several terms that you will need to always take into account.
- Then, the next sections will explain each of the typical sections of a technical paper: their structure, their objectives, and several general tips for them.
- After reviewing the different sections of a paper, we will also take a look at what's next: presenting your paper. Both in a public presentation and with a poster.
- Finally, we conclude with some conclusions on the most important concepts to extract and remember from these notes.

General concepts and tips for your whole paper

The best way to write papers is to read and analyze them. Read all the tips and strategies in this document and check if the papers that you read follow them and how. That way you will interiorize how to apply them.

Audience

Each section of your paper has a different audience, that is, a different kind of people that are expected to be able to completely understand or reproduce the content that you present in your paper.

Taking your audience into account is fundamental to write a coherent paper.

Flow

One of the most important things when writing papers is to ensure **flow**. That is, to make sure that each sentence is connected to a main idea. A trick to keep flow is to use the formula “This + a summary word”.

Consistency

Very related with flow is the idea of consistency. There is consistency in a paper if **words and ideas are presented in a similar and orderly fashion that is logical**. Consistency promotes harmony and clarity within the content by creating a logical flow of the material that is understandable from the reader's perspective

Writing longer sentences is good only if it improves consistence.

Verb tenses and voices

When presenting and discussing your results it is interesting to use **personal** language such as “we report the following results...”. It boosts confidence in the reader. You can keep using passives for the rest of the paper if you want. Typically, we want to avoid for the rest of the paper the human effect.

Similarly, when presenting your results, you need to use **simple past** tenses. Use present tenses only for accepted knowledge.

Tables

Also, do not re-explain everything that is on a **table**. Allow it to speak by itself and only highlight some important remarks from it. Keep this to an extreme. If you only need to say “(Figure 2)” instead of “as is shown in Figure 2” then so be it. Also, **captions of tables go above whereas captions of figures go below**.

Common typos

Let me address now the typical typos that many papers have. Try to be aware of them to avoid them!

- **Putting a gap for percentages:** don't! You do not put a gap for percentages. So instead of 98 % you write 98%.
- **Not capitalizing the first letter in tables and figures:** don't! When referring to figures and tables you always capitalize the first letter, as in “you can observe in Table 1 that...”.

- **Capitalizing the first letter of methods:** don't! Methods such as "reinforcement learning" or "adversarial networks" have to be written as any other word. Do not write "now, we will be using Reinforcement Learning for the following problem...". Instead, write "now, we will be using reinforcement learning for the following problem".
- **Using the Oxford comma wrong:** Be wary of the Oxford comma. It is generally expected in American English journals. It is the comma that you put after the last element of a list: "the table, the chair, and the lamp".
- **Starting a sentence with a symbol:** Just don't. This is not a valid sentence: "900 experiments were performed". Instead, write: "We performed 900 experiments". Do not write " α was set to a value of 0.5". Instead, write "the parameter α was set to a value of 0.5".
- **Avoiding all punctuation marks when using formulas:** don't! A mathematical equation in a paper, even if it gets its own spot alone with the $\begin{equation}...\end{equation}$, still has to respect all punctuation marks. In short, if after your equation you start a new paragraph, then you need to add a dot "." at the end of your equation, just like you would do with any other sentence with natural language.
- **Citing the wrong authors:** Just take into account that when you mention that something is by **John Smith et alii** it is because Smith is the standard or the pioneer of some technique, don't throw names at random.

Mastering the Abstract section

The **abstract** of your paper could be defined as a very concise summary of the whole paper. It is always an original work, not an extracted passage from your paper.

An abstract must be **fully self-contained and make sense by itself**, without further reference to outside sources or to the actual paper. It highlights key content areas, your research purpose, the relevance or importance of your work, and the main outcomes.

It is a well-developed single paragraph of less than 250 words in length, which is indented and single spaced. The function of the abstract is to outline briefly all parts of the paper.

Although it is placed at the beginning of your paper, immediately following the title, **the abstract should be the last thing that you write**, once you are sure of the conclusions you will reach.

What's the point of the abstract?

The abstract section is the first thing that anyone will read in your paper. It is your first (and in many cases your only) opportunity to let any researcher know, among other things:

1. The kind of problems you are dealing with.
2. Your proposed novel methodology.
3. The importance of your work.

An abstract lets a researcher quickly filter if your work is related with theirs or not. This becomes painfully clear if you think about how **you** use the abstract section of any paper that you read:

- Probably you are trying to look for papers related with yours, that tackle the same research problem or a similar one.
- So, you take a look at the abstract section of your pile of potential papers to read so you can discard those that are unrelated.

From those that seem interesting, which one are you going to read first? Suppose that you have two candidates:

- One that dully explains everything that was done, but without really putting into context the significance of their results.
- Another one that also states what was done, but also highlights the importance of their results, putting them into context.

You probably would select the second one.

So that's the whole thing. **The point of the abstract is to:**

1. So that people in the same research area can discover your paper.
2. And then, trying to convince them to read your full paper.
3. Also, they are mandatory on almost any conference or journal.

Organizing your abstract

An abstract is typically divided in **three moves**.

1. **State what was done:**
 - a. Identify the research area and its importance (optional).
 - b. Mention a gap addressed by the work (optional).
 - c. State the purpose and / or accomplishment(s) of this work.
2. **Identify the methods you used:**
 - a. Describe the methodology and / or procedures.
3. **Report your principal findings:**
 - a. Highlight major results
 - b. Offer a concluding remark

General tips for your abstract

Now I will provide several tips and “rules of thumb” when writing your abstract. Do not take them as set in stone, they are just a guide to make your writing process a bit easier.

1. **When:** your abstract section should be written last, once you have the rest of the paper finished. It is a concise summary of the rest of the paper, after all.
2. **Originality:** First of all, don't say “**nowadays**”, everybody uses that word. Especially don't use it as the first word of your abstract. Using the present tense is enough! Alternatively, you can use “increasingly”. In any case, being a bit original can create a good first impression with your abstract.
3. **Length:**
 - a. Regarding its length, an abstract section needs to be **250 words at maximum**.

- b. **What about the length of each sentence?** A **sentence** can be as long as it needs so long it remains coherent. However, it is eagerly encouraged to, if possible, express everything in short sentences to ease the readability of your abstract.
4. **Audience:** Always consider who is going to be your **audience, which is different for each section**. In the case of the abstract, the audience is everybody, including non-experts, as we have established. You have to write your abstract accordingly. Always think: **could my abstract be understood by a 14-year-old teenager?** Yes, that should be your goal.
5. **Reaching out:** Once your paper is published, the first thing that people will look at is your abstract. In fact, it is typically the only publicly accessible part of your paper in many occasions, so it is especially important to take into account:
 - a. **Search Engine Optimization:** one of the most important tools currently. At least give it a thought.
 - b. **An abstract is like a movie blurb:** it has to be intriguing; it has to make the reader want to know more about your work. That's the whole point of your abstract. Of course, it also needs to be informative enough.
 - c. **But always be honest!** One thing is to be enthusiastic about your results, a very different thing is to straight up lie about them and tell your audience a different thing that what they'll read in the full paper.
6. **Conciseness:** Yes, your abstract is very short, and it is impossible to explain all the work you did in only 180 words. So, the first thing is to accept that you cannot explain everything in detail in your abstract (you have the full paper for that after all!). Instead, you have to be:
 - a. **Concise:** which is different from being detailed! Highlight your major results (follow the previously presented 3-move structure for your abstract), but there's no need to enter into details here.
 - b. **Do not detail everything!** Let me repeat it to make it clear.
 - c. **You can take things for granted!** An abstract does not need too much signposting, you can go straight to the point.

Reviewing your abstract

You have finally written a first version of your abstract. Now it is the time to check everything that you have learnt and start working on reviewing your abstract to write an even better version.

When reviewing your abstract, ask yourself the following ten questions after writing it:

1. How well do you address an expert audience?
2. How easy is to follow your organization?
3. How clear and concise is the writing?
4. How does your abstract identify the research area and its importance?
5. How well does it identify the purpose of the study?
6. How well does it describe the methodology?
7. Does it include an effective concluding remark?
8. How appropriate is your use of writing conventions (e.g., tense and voice, paraphrasing)?
9. How accurate is your grammar and mechanics?
10. How accurate is the science content?

Even better, send your abstract to one of your colleagues and ask them to answer these 10 questions. This will give you a much more honest review.

Mastering the Introduction section

The introduction section offers a concise summary of the paper, slightly more detailed than the abstract. For a conference paper, it typically takes a whole page, which is approximately four times larger than the abstract.

In many ways, the introduction section could be summarized as “the abstract, but extended”. Do you remember all those details that you could not include in your abstract due to the word limit? The introduction is the perfect place for (some of) them.

Moreover, you will see now why having an excellently written introduction section is paramount. In many ways, the quality of your paper is 90% determined by the quality of your introduction.

What’s the point of the introduction?

Think of the introduction as the definitive test to convince a researcher to read your paper. You got their attention with the abstract, now you need to **convince** them to read the whole thing.

Again, we will make the same thought experiment. Think on your own approach to reading the introduction section of any paper.

Probably the abstract sounded interesting to you, they mentioned an interesting new method, or perhaps they offer novel results on a research area that is highly related with your own research. In any case, you want to know more.

So, you start reading their introduction. What do you expect to find here? Essentially, you want to know more details. More details about:

- The kind of problems they are dealing with (their research area).
- The methodology that they are applying (and how it compares with the rest of the community).
- The contributions of their work, and their significance.

You want to know more details, but **not all details**! You want them to do the hard work for you of summarizing the important parts of their work, because they know you are a very busy person.

An introduction that overstays its welcome defeats its own purpose.

But of course, you do not want them to be too succinct in the introduction. While the abstract is just an appetizer, the introduction should provide you enough **information to exactly know what to expect in the rest of the paper.**

And again, it’s all about marketing. There are many ways to explain the same results. But which paper would you read first? One where its introduction has a strong opening sentence, and an explanation on why their found results are so exciting? Or another paper that limits itself to be a list of what has been done, as if it was a bureaucratic requisite?

So that’s the whole thing. **The point of the introduction is:**

1. So that people in the same research area can get an idea of what to expect in the rest of your paper.
2. And then, trying to convince them to read your full paper.

The introduction will be your last opportunity! If a researcher is not convinced by your introduction, they will not even bother reading the rest of the paper. For that reason, **the introduction section is the most important section of your paper.**

Organizing your introduction

An introduction is typically divided into several moves:

1. Introduce the research area:
 - a. Identify the research area.
 - b. Establish the importance of the research area.
 - c. Provide essential background information about the research area.
2. Identify a gap (or gaps):
 - a. Indicate a gap in the previous research or raise a question about it. Here is when you quickly attack the related work.
3. Fill the gap:
 - a. Introduce the current work, your contributions.
 - b. Preview key findings of the current work (*optional*, depends on the research area).

Now let's talk in more detail about each move.

Introducing the research area

When introducing your research area, you need to cite and summarize others' work in concise and appropriate ways. Take a look at how detailed are the authors' descriptions of others' work in any paper that you are currently reading.

Always try to conform to the Instructions for Authors of the journal or conference you are submitting your paper. For example:

- The introduction should state the purpose of the investigation and must include appropriate citations of relevant, precedent work but should not include an extensive review of marginally related literature (*American Chemical Society*).
- Do not attempt a complete survey of the literature. In general, the introduction should be no more than two double-spaced pages without figures or tables and should include fewer than 20 references (*Env. Science and Tech*).

Identifying the gap

When reading a paper, if you want to identify the **gap**, search for: "Yet, ..."

There are many ways of identifying a gap. Here a few examples:

- A question that remains unanswered: "Numerous questions remain unanswered about [Y]".
- A research area that remains poorly understood: "Although much has been learned about [X], [Y] remains poorly understood.
- A next step that needs to be taken: "The next step is to apply [X] to the study of [Y].
- An area that has yet to be studied: "[X] has been the subject of several studies; however, to the best of our knowledge, no studies on [GAP] have been reported".

- A procedure that needs to be improved: “Although [X] achieves the desired results, the method is costly and time-consuming”.
- A new hypothesis that needs to be validated: “Additional studies are needed to corroborate these findings”.

Where [X] = “what has been done, learned, or understood” and [Y] = “what needs to be done, learned, or understood (the gap)”.

Filling the gap

When reading a paper, to identify how the authors **fill the gap**, search for: “Here, we...”. **The gap is filled in present simple**. Other options are:

- Here, we...
- In the present study / work, we...
- In this context / investigation / study / work / paper, we...
- Herein, we...

General tips for your introduction

Now I will provide several tips and “rules of thumb” when writing your abstract. Do not take them as set in stone, they are just a guide to make your writing process a bit easier.

- **Audience:** The audience to the introduction, similarly to the abstract section, is everybody, including non-experts.
- **Begin with a bang:** When writing the Introduction, it is very important to start with an **impactful opening sentence**.
- **Structure:** Each paragraph in the section should have a **major purpose** (following the move structure).
- **Related work:** In general, the Introduction **is not a complete survey** of the literature (that’s what the related work section is for). It must include appropriate citations of relevant, precedent work.
- **Purpose:** In the introduction, you need to tell the audience upfront what’s in it for them.
- **Tenses:** both present and past tenses are used in the Introduction.
 - o **Past tenses:** Generally, you use PAST tenses to explain what has been done so far (“some work has focused on...”).
 - o **Present tenses:** Generally, you use PRESENT tenses to explain how you fill the gap (“here, we show that...”), your experiments (“we compare our algorithm with the state-of-the-art”) and also common-knowledge facts (“convolutional neural networks provide spectacular results on computer vision tasks”).

Reviewing your introduction

You have finally written a first version of your introduction. Now it is the time to check everything that you have learnt and start working on reviewing your introduction to write an even better version.

When reviewing your introduction, ask yourself the following ten questions after writing an Introduction:

1. How accessible is your introduction to a broad audience? Is expert knowledge required to understand the Introduction?
2. How easy is to follow the organization of your Introduction? How closely do you follow the three-move structure?
3. How clear and concise is your writing?
4. How clearly and appropriately did you summarize and cite relevant literature?
5. How clearly and appropriately did you introduce the research area?
6. How clearly and appropriately did you identify a gap?
7. How well have you filled the gap?
8. How appropriate is your use of writing conventions (e.g., tense and voice, paraphrasing)?
9. How accurate is your grammar and mechanics?
10. How accurate is the science content?

Even better, send your introduction to one of your colleagues and ask them to answer these 10 questions. This will give you a much more honest review.

Mastering the Methodology section

The methodology section offers your contributions. Here is where you will explain the new methods / algorithms / processes that you have developed, and the necessary background information to understand them and why they are such an interesting and novel contribution. Of course, everything has to be backed up by a claim, so if your novelty is that it surpasses the performance of the state of the art, you will need to wait until the results section for that.

The methodology section is typically the longest section of a paper. In a conference paper, it may take between 4-5 pages approximately.

What's the point of the methodology?

The methods section is intended for an expert audience.

You need to be able to provide the necessary information so that an expert (not a novice) in the area will later be capable of reproducing your experiments only looking at what you have written and reported.

Taking that into account, imagine that you did your experiments with a novel kind of neural network, which slightly modifies convolutional neural networks, but has nothing to do with recurrent neural networks. Then...

- Would you need to provide **background information** about neural networks? Yes, but only the necessary bits.
- Would you need to provide **background information** about convolutional neural networks? Yes, but only the necessary bits.
- Would you need to provide **background information** about recurrent neural networks? No.

In essence, it all boils down to:

- Explaining the things that you needed to develop your methods.
- Explaining your novel methods.

In many occasions, if you see that you need to explain several bits of background information, it can be a good thing to separate them in their own background section (before the methodology section of course!), in which you only explain them but without detailing for what you use them (you will do that later in the methodology section). In this case, it is perfectly acceptable to explain the background in present tense, because they are all well established methods, and you are not really explaining (yet) how you used them.

Finally, by convention, the passive voice is typically used here.

Organizing your methodology

A methodology section is typically divided in several moves.

1. **(Optional) Describe the background information:** all theorems, definitions, and algorithms that you did not discover.
2. **Describe the kind of data you investigate:** in the case of reinforcement learning, this will be the family of learning environment of the agents. In essence, here you present the models that you are interested in studying.
3. **Describe your theoretical computations:** describe all the theoretical analyses that you have performed to the data you investigate. For example, if you have proved an interesting novel theorem relating this data with a specific family of algorithms.
4. **Describe your algorithms:** describe all the procedures that you are interested in applying to the data you investigate. For example, if you have designed a novel algorithm, this is the time to explain it.

Reviewing your methodology

You have finally written a first version of your methodology section. Now it is the time to check everything that you have learnt and start working on reviewing your section to write an even better version.

When reviewing your methodology section, ask yourself the following ten questions after writing it:

1. How well do you address an expert audience?
2. How easy is to follow the organization of your draft?
3. How clear and concise is the writing?
4. How well does the draft describe the data used?
5. How well does the draft describe computational procedures and / or the instrumentation? (e.g., the computational power needed)
6. How well does the draft describe theoretical procedures?

7. How appropriate is your use of writing conventions? (e.g., capitalization, abbreviations and acronyms with parentheses and spelled out terms, numbers and units, past and present tense, active and passive voice)
8. How accurate is your grammar and mechanics?
9. How accurate is the science content?

Even better, send your methodology section to one of your colleagues and ask them to answer these 10 questions. This will give you a much more honest review.

Experiments section

The experiments section is intended for an expert audience.

Similarly to the methodology section, you need to be able to provide the necessary information so that an expert (not a novice) in the area will later be capable of reproducing your experiments only looking at what you have written and reported.

The scope of your experiments will determine the scope of your contributions and in turn the scope of the whole paper. Thus, it is not only about obtaining great results from the experiments: you also need to carefully design experiments that make sense. 90% of the times, the excuse for rejecting a paper is that “its experiments were not significant enough”. Don’t give them that excuse!

What’s the point of the experiments?

In the introduction you explained everything about your novel method / algorithm / whatever that greatly contributes to the state of the art. Then, in the methodology you explained what it was and how it works. The experiments section is now the time to put into test that your method is a significant contribution. Afterwards, in the results section, we will check if it was a contribution good enough or not.

The experiments section and the results section may or may not be fused into a single section. Here, for simplicity, we consider them two separate questions.

In essence, it all boils down to:

- Reporting what kind of experiments you did detailing so that someone else can reproduce it.
- Explaining why you chose these specific experiments. What were you trying to prove? What was your hypothesis?
- Choose the necessary experiments to later prove that your contribution is significant!

The verb tense for the experiments section is the **simple past** because you HAVE ALREADY DONE all your experiments. You are not providing a cookbook with step-by-step instructions on how to imitate your experiments. You are explaining WHAT YOU DID. Let me rephrase it: DO NOT USE THE FUTURE TENSE. A paper is not a transcript of a live show.

Organizing your experiments

An experiments section is typically divided in several moves.

1. **Describe the experiments that you performed:** In general, for a machine learning algorithm, this means explaining the training and testing data sets, and the hyperparameters of your learning algorithm.
2. **Describe your evaluation metrics:** these are the metrics that will later help us discern if your proposed method / algorithm / process is truly significant or not.
3. **Describe your competitors:** To be significant means to do something better than the state of the art (and by a wide enough margin). You need to explain which methods / algorithms / processes are your competitors in the state of the art (to which you will apply exactly the same experiments and metrics that you applied to your own contribution). Even if there is no competitor due to being a very new area, you at least need to say it and provide good enough evidence!

Reviewing your experiments

You have finally written a first version of your experiments section. Now it is the time to check everything that you have learnt and start working on reviewing your section to write an even better version.

When reviewing your experiments section, ask yourself the following ten questions after writing it:

1. How well do you address an expert audience?
2. How easy is to follow the organization of your draft?
3. How clear and concise is the writing?
4. How well does the draft describe the experiments applied? Could an expert reproduce them?
5. How well does the draft describe the metrics applied? Would an expert understand them?
6. If they were successful, would these experiments allow you to back up your claims in the introduction?
7. Are these experiments sufficient to put into test all your claimed contributions?
8. How appropriate is your use of writing conventions? (e.g., capitalization, abbreviations and acronyms with parentheses and spelled out terms, numbers and units, past and present tense, active and passive voice)
9. How accurate is your grammar and mechanics?
10. How accurate is the science content?

Even better, send your experiments section to one of your colleagues and ask them to answer these 10 questions. This will give you a much more honest review.

Results section

While the methodology section and the experiments sections were intended for an expert audience (after all, no one else would be able to replicate your experiments), the results, and what's left of the paper, should be oriented for a much wider audience. Probably not as wide as the introduction audience, but definitely wider.

Remember: the more clearly and simply you explain your results, the stronger will be your later arguments to prove that your contributions are significant. If you explained a metric in your

experiments section such that “number smaller than 0.05 good”, “number greater than 0.05 bad”, this is something that even a 13-year-old student can understand. Of course, YOU need to make that explicit.

Now, some general comments:

- Don't say something is **true or false** NEVER in the discussion section or in the results section EXCEPT if you are absolutely certain and have irrefutable proofs.
- For explaining your results, never use the word “**done**” or the verb “to do” in general. Use “perform” instead.
- **Don't hedge results.** Results happened!! So, no need to hedge them. What you need to hedge is your **interpretation** of them (in the discussion).
- If something is **general well-known information** you don't need to put a citation for it.
- A **13-year-old** should understand your sentences. Always make everything simpler.

What's the point of the results?

In the experiments section you explained how you were planning on evaluating your novel contribution. Now, in the results section, we will check the numbers you obtained and help the reader to understand them.

The experiments section and the results section may or may not be fused into a single section. Here, for simplicity, we consider them two separate questions.

In essence, here is where you tell the end of the story. You discovered a shiny new toy (methodology), you decided to play a little bit with (experiments), and now you present what happened. Always remember this flow: there is a clear connection between these sections. The more explicit you show this connection, the more readable the paper will be.

Results are written in **simple past**. Notice that even if results are written in simple past, then when we move to their interpretation (discussion) everything becomes now **simple present**. The summary is also in present.

Organizing your results

A results section is typically divided in several moves that can be repeated for each experiment that you explained in the experiments section.

1. **Remind what were you trying to prove with this experiment:** That is, your contribution. We want your reader to always bear it in mind so that they can understand what is the meaning of the obtained results.
2. **Explain as simple as possible how to interpret your results:** This means explaining how to interpret the metric you explained in the experiments sections. This is the part that needs to be understandable by a 13-year-old. If you already did this in the experiments sections then you do not need to re-explain everything again here, but at least give some pointers to refresh the reader's memory.
3. **Describe the obtained results:** In terms of the metrics you explained in the previous section. If you have tables or figures to better understand these metrics, you have to explain them here. If you have a separate discussion section, do not comment anything subjective about the obtained results. When describing your results, we expect to see at least one of these three:

- a. **Important findings:** you can retroactively modify your experiments section if some of your findings are not very significant nor exciting. The amount of time of a reader is limited, so stick to the most shocking results.
 - b. **Trends:** If your results are not revolutionary, it is fine if at least they show a trend. For instance, you proved that an hyperparameter of a well-known algorithm and a given metric are directly proportional.
 - c. **Unexpected results:** something that clearly goes against your hypotheses or what would be expected by the research community.
4. **Objectively compare your results.** If you performed 10% better than the state of the art, this is not subjective: you can include it here. If your experiment undoubtedly proves your hypothesis, you can also comment it here. If it is dubious, you will need to wait until the discussion section.

Reviewing your results

You have finally written a first version of your experiments section. Now it is the time to check everything that you have learnt and start working on reviewing your section to write an even better version.

When reviewing your experiments section, ask yourself the following eleven questions after writing it:

1. How well do you address an expert audience?
2. How easy is to follow the organization of your draft?
3. Did you include at least one of the following: important findings, trends, and/or unexpected results?
4. How clear and concise is your writing?
5. Did you remember to remind the reader of the point of your experiments? How clear and appropriate is that reminder?
6. How clearly and appropriately have you referred the reader to each figure and table?
7. How well have you told the story of scientific discovery of your results? (i.e., don't be boring)
8. To what extend have you formatted graphics clearly and correctly?
9. How appropriate is your use of writing conventions? (e.g., capitalization, abbreviations and acronyms with parentheses and spelled out terms, numbers and units, past and present tense, active and passive voice)
10. How accurate is your grammar and mechanics?
11. How accurate is the science content?

Even better, send your results section to one of your colleagues and ask them to answer these 10 questions. This will give you a much more honest review.

Discussion section

The discussion section is a relatively small section in which you go back at your results and try to make sense of them. Here is when you explain with your best arguments how they prove the contributions you promised at the introduction.

Of course, there is always room for alternative interpretations of results, that's fine! But let the reviewer do the pessimistic interpretation. You will be the only one interpreting them in a good light. Nobody else will because

- a) They will not have time to process and analyze your results (you have to do that for them).
- b) By default, everyone assumes that your results are not significant. You have to defend yourself!

Sometimes, the discussion section is simply a subsection of the experiments section. That is completely fine, but its structure remains the same.

Some papers completely fuse both sections and explain them together. This is fine as long as it will be perfectly clear what are your results and what is your discussion for each result. Remember that your goal is to always explain everything in the clearest way.

What's the point of the discussion?

Similarly to a criminal judgment, the discussion section is the time for your final allegations. We have seen all the proofs at the results section, now is your time to be your own lawyer and defend what you got as much as possible.

In a sense, the discussion section is all or nothing. You have to explain here how your results back up your contribution claims, and this is your only chance.

Organizing your discussion

A discussion section is typically divided in several moves.

- First, you remind readers of your results.
- Then, you interpret them. You hedge finds and cite the relevant literature to compare them.

Common hedging words include:

- Apparently, appear, can, could, generally, indicate, largely, likely, mainly, may, might, possible, possibly, potentially, presumably, probable, probably, seem, should, suggest, support, typically, would.

Reviewing your discussion

You have finally written a first version of your discussion section. Now it is the time to check everything that you have learnt and start working on reviewing your section to write an even better version.

When reviewing your discussion section, ask yourself the following eleven questions after writing it:

1. How well do you address an expert audience?
2. How easy is to follow the organization of your draft?
3. How clear and concise is your writing?
4. Did you hedge your results? Did it become clear that you are hedging them and not providing false claims?
5. Does your discussion puts into context the obtained results?

6. Does your discussion confirm the contributions that you stated at the introduction?
7. Does your discussion include anything that could be used by the reviewers to completely destroy you? Then remove it, or if it cannot be removed it, try to explain it as positively as possible without lying.
8. How appropriate is your use of writing conventions? (e.g., capitalization, abbreviations and acronyms with parentheses and spelled out terms, numbers and units, past and present tense, active and passive voice)
9. How accurate is your grammar and mechanics?
10. How accurate is the science content?

Even better, send your discussion section to one of your colleagues and ask them to answer these 10 questions. This will give you a much more honest review.

Conclusions section

The conclusions section is the smallest one. It typically has a size similar to your abstract. Like in a song, this is where you end with a bang. You have just been discussing your results in detail, but what are the final messages and takeaways from your paper? This is the place to make them explicit.

These final messages are simply the contributions of your work, which now that you have had your discussions section you can back up with evidences.

What's the point of the conclusions?

The **purpose** of this section is twofold.

- First, you summarize again the main contributions of this work so that a read can have a clear idea of what is the novelty here and what is its significance.
- Then, you add a **future work** part, whose purpose is to do damage control. There you acknowledge things that are important, but you did not do.

The **audience** of the conclusions and future work section is a person that has already read your whole paper and followed through it. You can now use all the technicalities to explain your contributions that you could not use in the Introduction section. However, remember that it should also be understood by a non-expert, so do not take it to an extreme.

Organizing your conclusions

A conclusions section is typically divided in several moves.

1. You repeat your main contributions so that at the end the reader has a clear idea of what were you trying to do.
2. You provide future work pointers, without entering into much detail. It is particularly important that you avoid sentences like “in future work we will need to compare our results with this other metric to actually verify its significance”. Please avoid attacking yourself (this is the job of the reviewers).

Reviewing your conclusions

You have finally written a first version of your conclusions on section. Now it is the time to check everything that you have learnt and start working on reviewing your section to write an even better version.

When reviewing your conclusions section, ask yourself the following ten questions after writing it:

1. How well do you address an expert audience?
2. How easy is to follow the organization of your draft?
3. How clear and concise is your writing?
4. Did you make explicit all your contributions here?
5. Did you make use of what you established in the results and discussion section here to put into context the significance and novelty of your results?
6. Do your conclusions match what was stated at the introduction?
7. Did you include at least two future work directions as damage control?
8. How appropriate is your use of writing conventions? (e.g., capitalization, abbreviations and acronyms with parentheses and spelled out terms, numbers and units, past and present tense, active and passive voice)
9. How accurate is your grammar and mechanics?
10. How accurate is the science content?

Even better, send your conclusions section to one of your colleagues and ask them to answer these 10 questions. This will give you a much more honest review.

Presenting your paper

Presenting your paper, either in poster format or in an oral presentation is typically the most terrifying part of the work. The panic comes from the fact that you will be directly confronting your ideas to a group of anonymous people.

But realistically, what's the worst thing that could happen? That the audience does not understand you and leave the room with the same knowledge from your work as before, but with even less interest in your work.

And realistically, what's the best thing that could happen? In the best scenario, most of the audience will understand the general ideas of your work, realize that is unrelated with theirs, and lose interest. But a few people will discover that you are working in a similar area, and they will be willing to read your papers or even start collaborating with you.

This directs us to the ultimate question of presenting your paper. **What's the point of presenting your paper?**

- **Is it to prove that your results in your paper are true?** NO! That was already assessed by your paper reviewers.
- **Is it to prove that you are an expert on the topic?** NO! That is already a given.

- *Is it to explain your audience how to replicate your results?* NO! Most of your audience will not be experts on the topic, so it would be impossible for them to follow your presentation, but most importantly, why should they?
- **Is it to promote your work and tell everyone how cool and interesting are your research contributions?** YES! That's the whole point, you are given a chance to promote yourself and your research line.

An oral presentation is an opportunity to explain to other people in 10-20 what are the cool results you found out in the paper, and what's their significance. In short, it is all about explaining why you did all this work.

A poster presentation is a similar idea. Think about film posters. They do not actually explain much. They tell you the main actors, who's the director, a picture probably showing a cool scene from the film (from which you can infer what's about) and that's it. They do not tell you the whole story.

An ideal poster condenses in a slide the cool results of the paper and what're their significance, so that people passing by will make a quick read of it and want to ask you more about it.

Preparing the slides of your presentation

The slides are a COMPANION to your oral presentation, not the other way around. Their only purpose is to help the audience follow your talk. That means that you must avoid putting anything too distracting in your slides. That means:

- **Don't put a lot of text in your slides.** It is preferable to have small very little text and some companion images.
- But **don't put either a lot of images on a single slide!** Similarly to how you will only be talking about one thing at a time, showing one image at a time is always the best option.
- We are getting repetitive here, **don't put a lot of tables on a single slide.**
- Finally, **put only the minimum amount of mathematical jargon** (including equations, formulas, numbers...) in your presentation. Ideally, you should be able to prepare your slides without a single equation.

Preparing your poster

When preparing a poster, remember that its goal is to maximize the amount of insight (not raw information) transferred to the reader and to generate a conversation. Design a simple poster with a main window that focuses on the main findings, and an ammo bar that describes your methodology and experiments in short sentences.

Preparing your oral presentation

When presenting your work, you need to tell the audience upfront what's in it for them in the first 2 minutes.

When doing a presentation, use fewer words and avoid complicated ones (such as "complicated").

Now, regarding phonics. English is a **STRESS** focused language. So, when talking, you need to stress the important words of a sentence, and not give the same strength to every single word.

Also, the strong syllable of a word always has a pure sound. And you should both slow it down for the strong syllable and say it louder.

Slow down when pronouncing important words in your work (like your new algorithm or the methodology that you use). Otherwise, they will not know what you're talking about.

When looking at an **elevator pitch** / abstract video: look at its organization, its strategies, and its individual words/phrases.

Last remarks

Finalizing your written work

- Identify the different features of effective writing.
- Examine your written work from various perspectives.
- Take systematic steps to improve the quality of your writing.
- Finalize your written work so that meets the expectations of your target audience.

Checklist

- **Check organization** - examine headings, subheadings, opening sentences in key paragraphs - this is a good indication that your broad organizational structure is sound.
- **Verify follow through** - Introduction typically foreshadows the rest of the work - "In this work, we demonstrate...". Do you overstate your case? Do you deliver on promises?
- **Eliminate redundancy** - For example: **Conclusion**. In conclusion...; in transitional moves when reminding reader of methods/results; in overlapping sections - e.g. the Abstract - try to state the information differently.
- **Maximize conciseness** - eliminate unnecessary words/delete information that can be assumed e.g. In our judgment/use nominalizations - after distillation vs. after we distilled the product/ focus on science by omitting authors' names
- **Check tense and voice** - when active make sure you used "we" sparingly and in conventional ways to signal decision or the start of the fill-the- gap statement.
- **Correct common errors** – acronyms + abbreviations/capitalization/numbers+ units/parallelism in series and lists/placement of figure + table captions/plural words/punctuation/spelling/s-v agreement.
- **Review science content** - high-quality writing and high-quality science go hand in hand.
 - Description of importance/implications
 - References to others' works.
 - Presentation of methods - check accuracy, written for expert
 - Presentation of results-logical story of scientific discovery
 - Interpretation of findings sufficient support and hedging
- **Reconfirm adherence to external guidelines** - read journal's Instructions for Authors.
- **Examine formatting and overall appearance** - 100% consistent.